ENGINEERING PROJECT PORTFOLIO

CHIMWEMWE CHINKUYU

B.S.E. Mechanical & Aerospace Engineering · Minor in Computer Science · Princeton University, Class of 2026

Chimwemwe Chinkuyu

EDUCATION

B.S.E. Mechanical & Aerospace Engineering · Minor in Computer Science · Princeton University, Class of 2026

MOTIVATION

I'm inspired by the possibility to turn ideas into smart, functional solutions at the intersection of hardware and software. My work spans mechanical design, fabrication, and software development.

I'm eager to grow in innovation-focused environments and contribute to advanced devices, autonomous systems, and technologies that enhance human health and quality of life.

TECHNICAL SKILLS

CAD / CAM
- PTC Creo (incl. FEA)
- Fusion 360
- Onshape
- AutoCAD

Programming & Robotics
- Python
- MATLAB
- Java
- C
- ROS
- COMPAS FAB
- UR-RTDE
- PID control

Fabrication
- 3D Printing (FDM, SLA, SLS)
- Manual & CNC Machining
- Carpentry
- Robotic Fabrication
- Autonomous Assembly

Software & Tools
- Simulink
- Jupyter Notebook
- Docker
- Anaconda
- Grasshopper
- NVIDIA Jetson
- RadiAnt DICOM Viewer

LINKEDIN linkedin.com/in/chimwemwe-chinkuyu

PORTFOLIO WEBSITE ckcgithub16.github.io/CC_Portfolio/

MECHANICAL ENGINEERING · SOFTWARE DEVELOPMENT

Autonomous Robotic Assembly of Multi-Layer Lincoln Log Structures

Sep 2025 – Apr 2026

UR3 arm assembling a Lincoln Log structure

OVERVIEW

My senior thesis asks a single question: how can an automated robotic system reliably assemble multi-layer Lincoln Log structures layer-by-layer with high precision? To answer it, I designed and built an autonomous system that uses a UR3 robotic arm to assemble these structures with no cameras, force sensors, or runtime feedback. It is the autonomous foundation for a longer-term goal of human-robot collaborative construction: before a robot can share the task with a person, it has to perform its own half reliably.

FOUR SUBSYSTEMS

The system integrates a parametric toolpath generator, a Cartesian motion planner, a synchronized execution controller, and a set of custom 3D-printed fixtures. The next two pages walk through the software pipeline and then the mechanical hardware and test results.

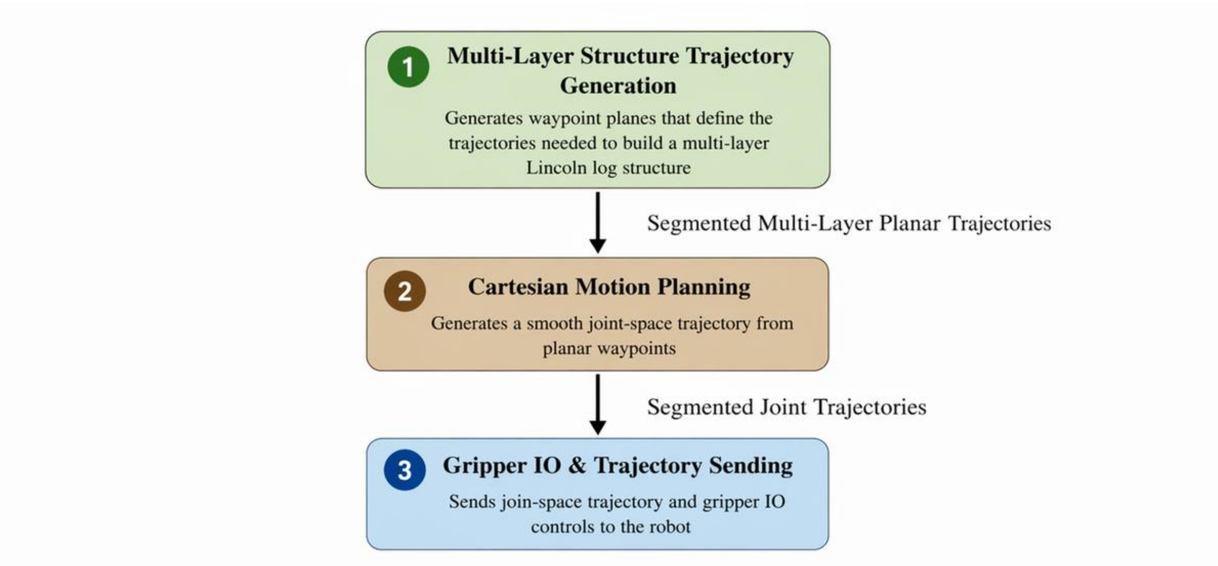
SKILLS USED

Python · Grasshopper / GhPython · COMPAS FAB · ROS · MoveIt! · UR3 · Fusion 360 · 3D Printing · Robotic Fabrication

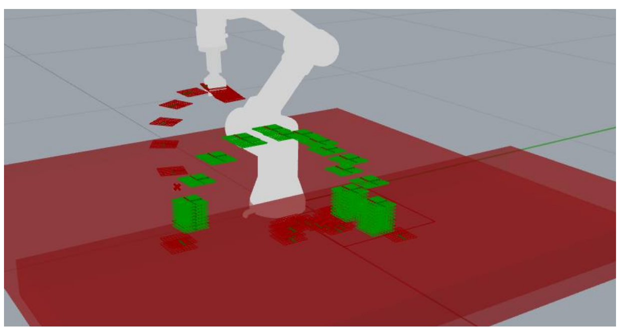
MECHANICAL ENGINEERING · SOFTWARE DEVELOPMENT

Autonomous Lincoln Log Assembly — Software Pipeline

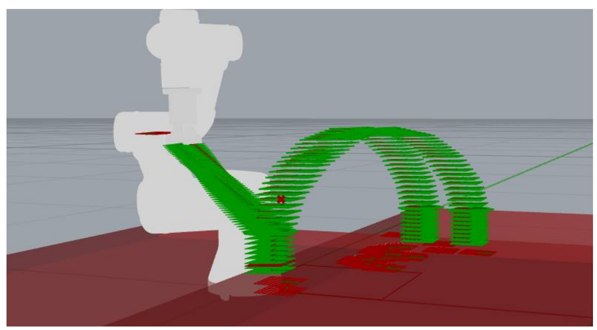
Sep 2025 – Apr 2026



Software pipeline — three GhPython components



Generated waypoint planes (Visualization in Rhino)



Planned Cartesian trajectory (Visualization in Rhino)

SOFTWARE PIPELINE

Three independent GhPython components in Grasshopper — linked through native DataTrees — generate geometry, plan motion, and drive the robot. Every log follows the same seven-waypoint path, so the gripper enters and exits along the z-axis.

PARAMETRIC TOOLPATH GENERATOR

Turns a few geometric inputs (pick and place planes, layer count) into the end-effector frames for every log, handling alternating 90° layers with strictly vertical approach and departure.

CARTESIAN MOTION PLANNER

Iterates over the waypoints and calls COMPAS FAB's plan_cartesian_motion for each segment, with MoveIt! collision checking. Cartesian planning keeps the gripper on a controlled arc that clears placed logs; every trajectory is previewed in Rhino first.

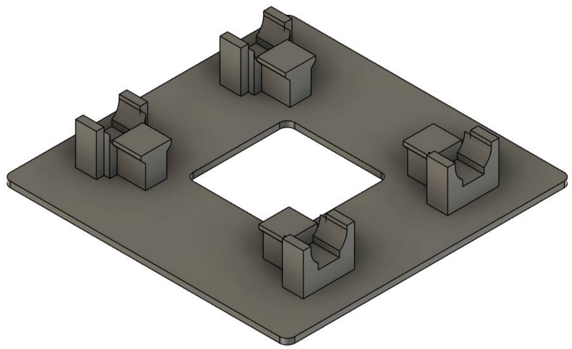
EXECUTION CONTROLLER

Streams the trajectories to the UR3 over one persistent RPC connection and fires the pneumatic gripper at exact segment boundaries — closing at the dispenser, opening at the notch, each with a 500 ms dwell.

MECHANICAL ENGINEERING · SOFTWARE DEVELOPMENT

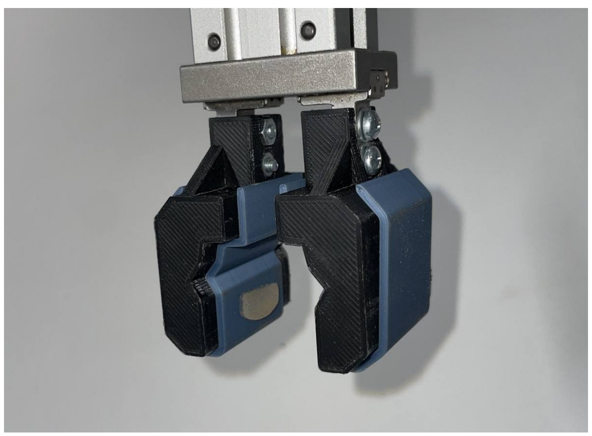
Autonomous Lincoln Log Assembly — Mechanical Design & Results

Sep 2025 – Apr 2026

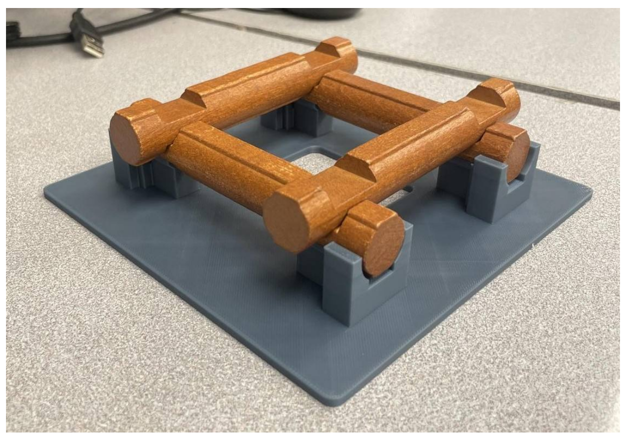


Calibrated structural base

Gravity-fed log dispenser



Gripper with custom covers



Completed two-layer assembly

MECHANICAL DESIGN

Three custom 3D-printed fixtures (Fusion 360 / Rhino, PLA) replace runtime sensing, so the robot never has to find or measure anything.

CUSTOM FIXTURES

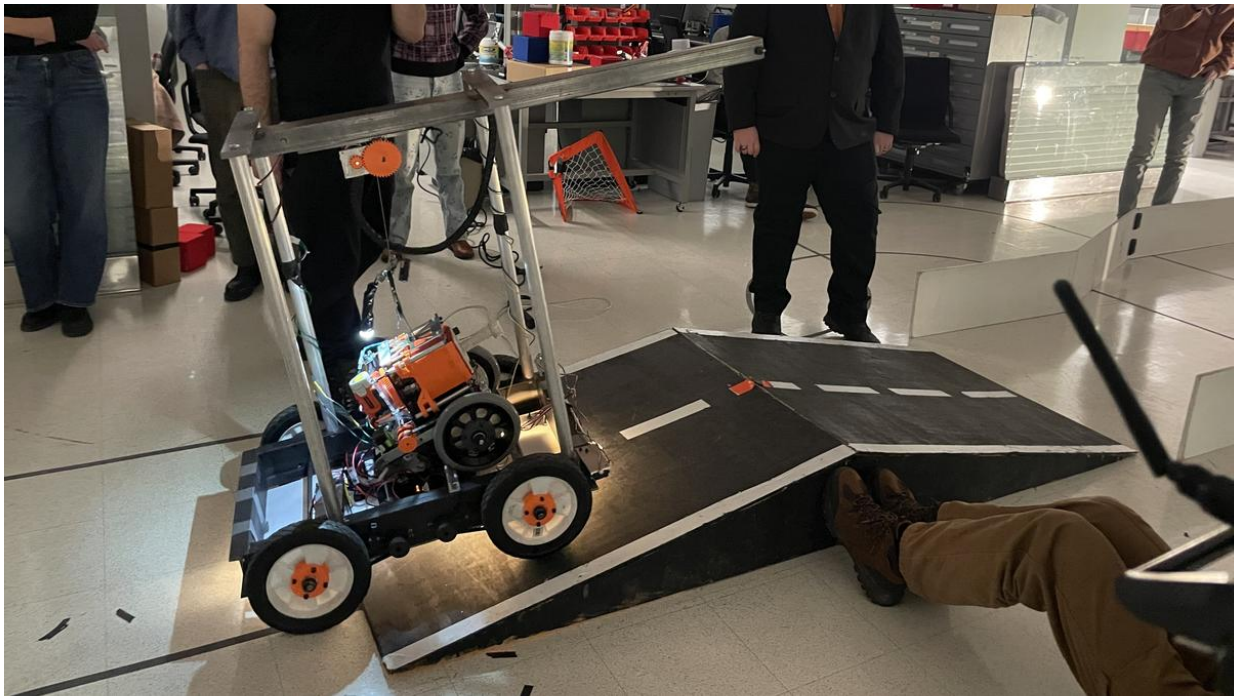
- A gravity-fed dispenser presents every log at the same pose from a 15° inclined channel — chosen over a spring magazine that analysis ruled out.
- A calibrated base fixes the first layer on exact 76.3 mm centerlines, so the system skips first-layer calibration.
- Custom gripper covers close the finger gap to log diameter for a positive-stop grasp.

TESTING & RESULTS

- Single-, two-, and three-layer builds, five trials each: 100%, ~75%, and ~83% success.
- Pick success was 100% every trial, validating the dispenser; all failures were placements.
- Failures were systematic: an odd-layer gripper-cover offset and an RPC connection-pool exhaustion at the 6th log — both diagnosed and fixed.

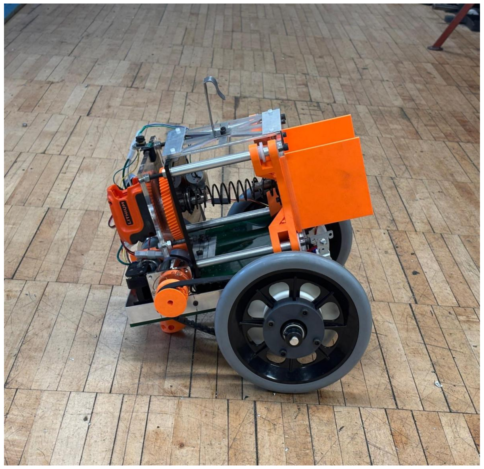
Design & Manufacturing of a Search and Rescue Robot

Oct – Dec 2025



SARR traveling over the ramp

Motherbot deploying Babybot over the wall



The Babybot

OVERVIEW

For a mechanical design course, I collaborated with 11 classmates to build a marsupial search and rescue robot that could navigate an obstacle course to deliver a med kit. The system pairs a large Motherbot that carries a smaller Babybot onboard. The Motherbot traversed a ramp, a chute, and a 36-inch wall. At the wall, a motorized lifting arm raised the Babybot over the top, deploying it on the far side. The Babybot then navigated independently to a target zone and delivered a med kit payload.

SENSING & CONTROL

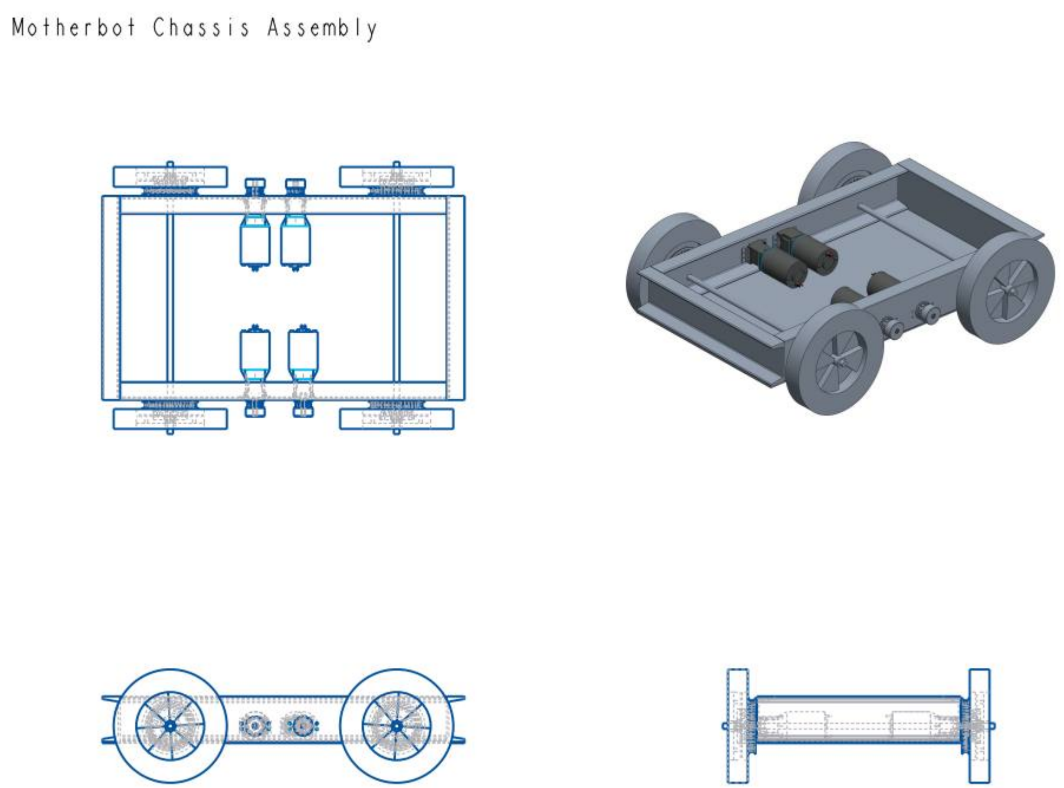
- Both robots used C++ state machine controllers on Teensy microcontrollers.
- The Motherbot used CdS phototransistors for line-following — proportional on flat ground, bang-bang on the ramp, PD in the chute — and ultrasonic sensors to trigger state transitions.
- The arm deployment was open-loop.
- The Babybot used similar sensing for post-deployment navigation.

RESULT

- The Motherbot navigated the entire course, deployed the Babybot over the 36-inch wall, and the Babybot delivered the med kit to the target zone at the class final.

Search and Rescue Robot — Chassis Design & Manufacturing

Oct – Dec 2025



MY CONTRIBUTION

- Designed the low-carbon steel chassis, 60:1 two-stage HTD belt drivetrain, and engineering drawings in Creo Parametric.
- Led fabrication of the chassis and drivetrain.

CHASSIS DESIGN

- AISI 1018 steel was chosen for weldability and stiffness.
- FBD analysis confirmed tip-over stability under full arm extension.
- FEA at motor mount positions and the lifting arm pivot verified structural margins.
- The two-stage belt drivetrain over a jackshaft provided torque for the ramp; slotted motor mount holes allowed precise belt tensioning.

MANUFACTURING

- Steel frame members were band-saw cut, cope-cut on a knee mill, and MIG welded.
- Motor mount brackets and the jackshaft were manually machined on the lathe and knee mill.
- Sensor mounts and the lifting arm cradle were FDM 3D printed for rapid iteration.
- Key drivetrain components were CNC machined for dimensional accuracy.

Custom Differential Gearbox

June 2025

OVERVIEW

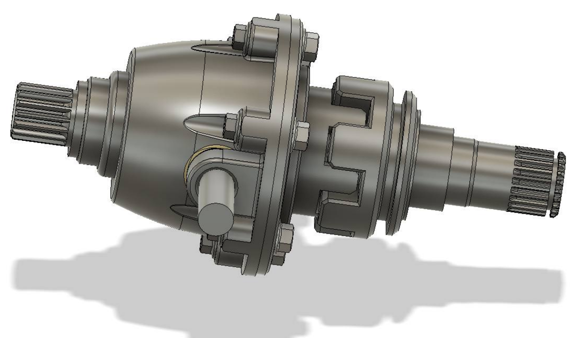
As a personal project to sharpen my CAD skills, I designed a fully parametric bevel-gear axle differential entirely in Autodesk Fusion 360. The planetary differential configuration distributes drive torque between two output shafts while allowing independent rotation speeds, enabling smooth, skid-free turning.

DESIGN APPROACH

- Gear geometry was determined using standard bevel gear formulas driven by linked Fusion 360 equations.
- Module, pressure angle, tooth count, and pitch circle diameter update the full geometry automatically as inputs change.

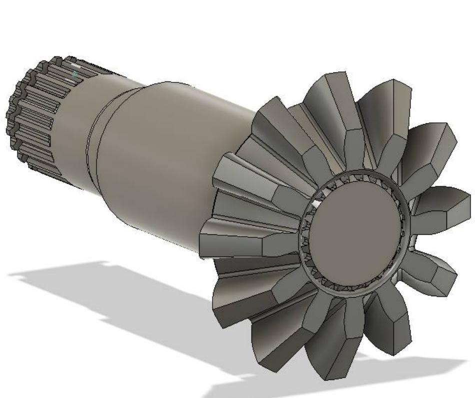
SKILLS USED

Autodesk Fusion 360 · Parametric Modeling · Bevel Gear Theory

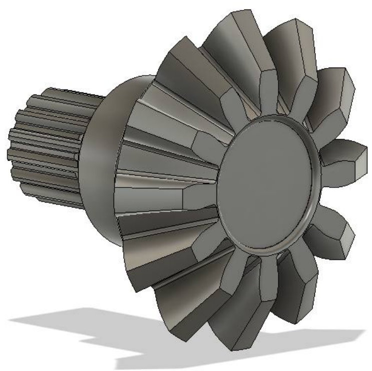


Custom Differential Gearbox — Component Drawings

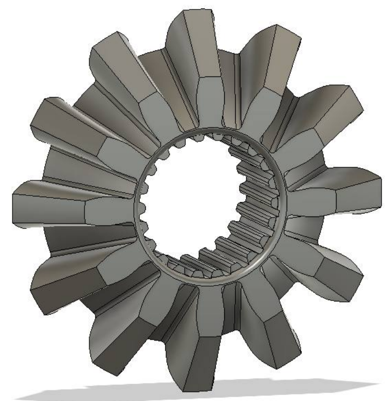
June 2025



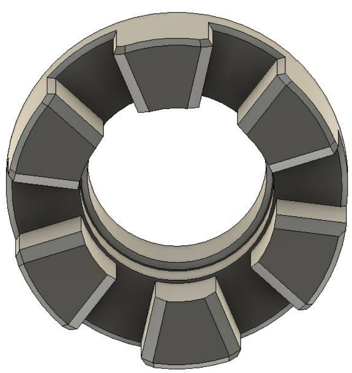
Long Drive Shaft



Short Drive Shaft

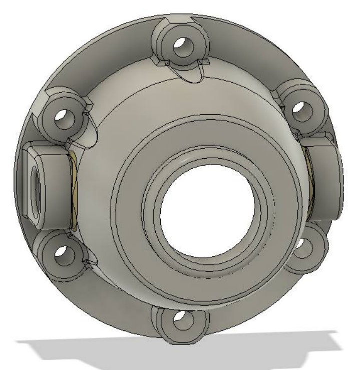


Bevel Gear



Differential Ring

Output Hub



Outer Casing

MECHANICAL ENGINEERING · SOFTWARE DEVELOPMENT

Self-Driving RC Truck with Obstacle Avoidance & Safety Filter

Mar – May 2025

Navigating the Task 1 obstacle course

Watch the team's Task 1 run on YouTube

OVERVIEW

For an intelligent robotics course, I worked in a team of 4 to program an autonomous RC truck, powered by an NVIDIA Jetson, to complete two tasks: racing through an obstacle course, and running a real-time driver-assist safety filter.

MY CONTRIBUTION

Algorithm development · Code implementation · Debugging of both software and hardware

TASK 1 — OBSTACLE COURSE RACING

- Implemented an iLQR (iterative Linear Quadratic Regulator) controller with cost-prioritization to navigate 16 ordered waypoints in 3:30 while avoiding up to 20 obstacle cubes.
- Finished 4th fastest of 14 teams after time penalties.

TASK 2 — SAFETY FILTER

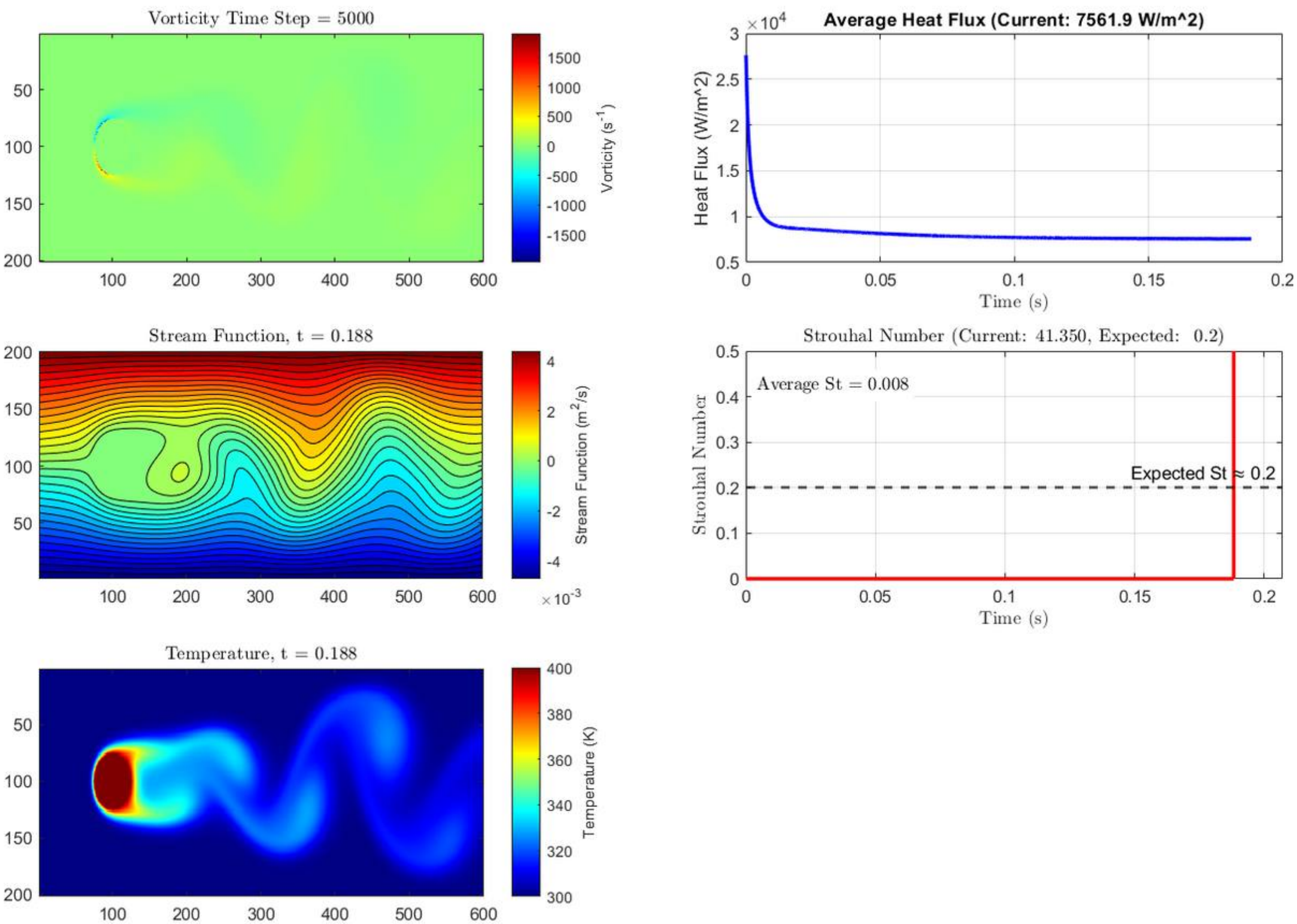
- Built an Advanced Driver Assistance System (ADAS) that uses iLQR to monitor manual driver inputs in real time.
- Overrides steering that would leave the track or cross lane boundaries while minimizing unnecessary interventions.

SKILLS USED

Python · ROS · Optimal Control · Path Planning · Obstacle Avoidance

2D Unsteady Heat Transfer: Heated Cylinder & NACA 2412 Airfoil

Apr – May 2025



Heated cylinder — vorticity, stream function, temperature, and convergence history

OVERVIEW

For a heat transfer course, I built a Psi-Omega (streamfunction-vorticity) finite-difference simulation from scratch in MATLAB to model 2D unsteady heat transfer over two geometries: a 400°C cylinder in 300°C flow (convective heat flux 7.6 kW/m²), and a NACA 2412 airfoil at 5° and 30° angles of attack (heat flux 1,002.6 vs 1,133.6 W/m²).

TECHNICAL APPROACH

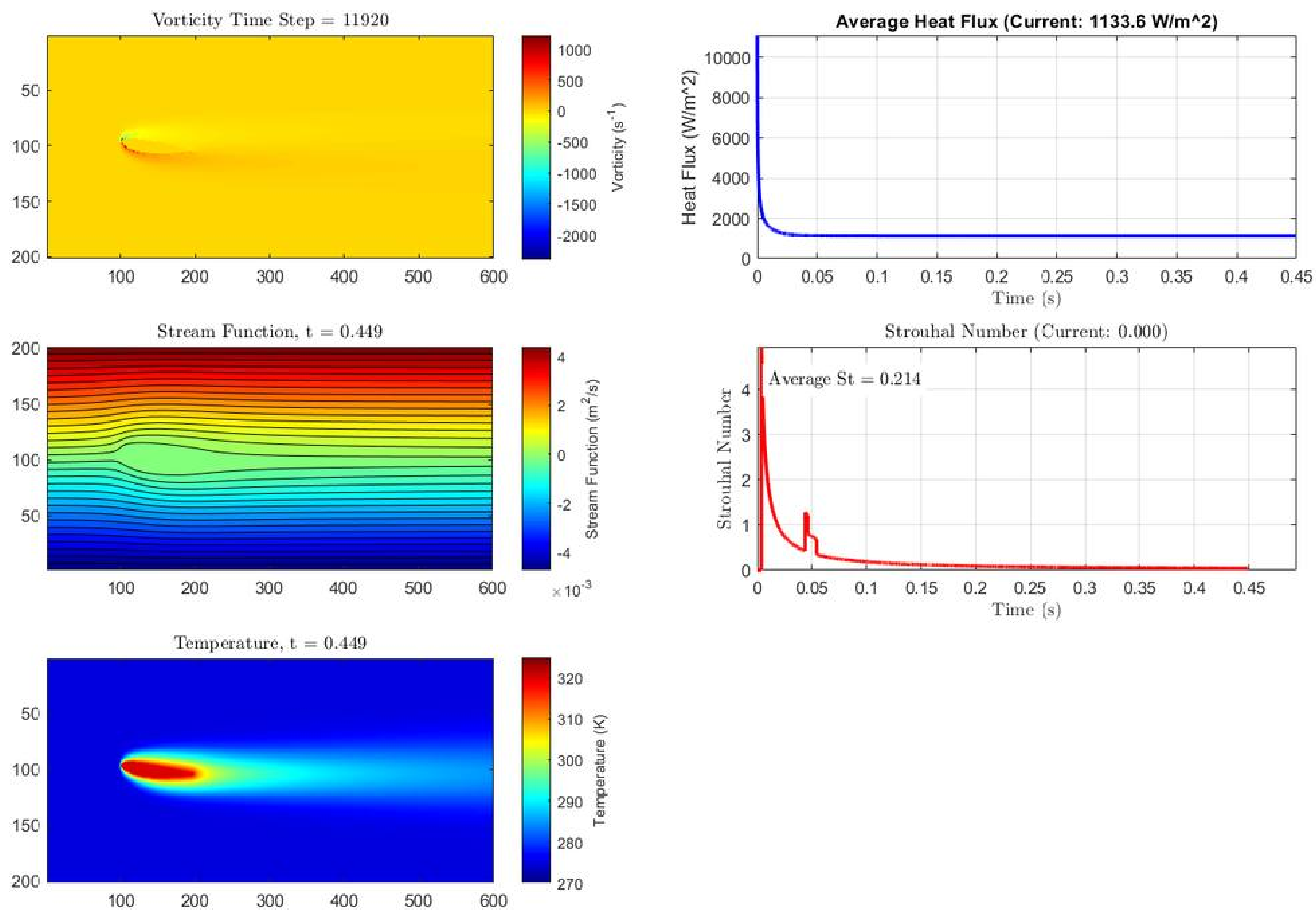
- Three coupled PDEs solved at each time step:
- Streamfunction (Poisson) via Gauss-Seidel with over-relaxation
- Vorticity (momentum) via forward Euler with upwind differencing
- Temperature (energy) coupled to the velocity field

SKILLS USED

MATLAB · Finite Difference Methods · Fluid Dynamics · Heat Transfer · Numerical Stability · CFD Validation

Heat Transfer Simulation — NACA 2412 at 5° Angle of Attack

Apr – May 2025

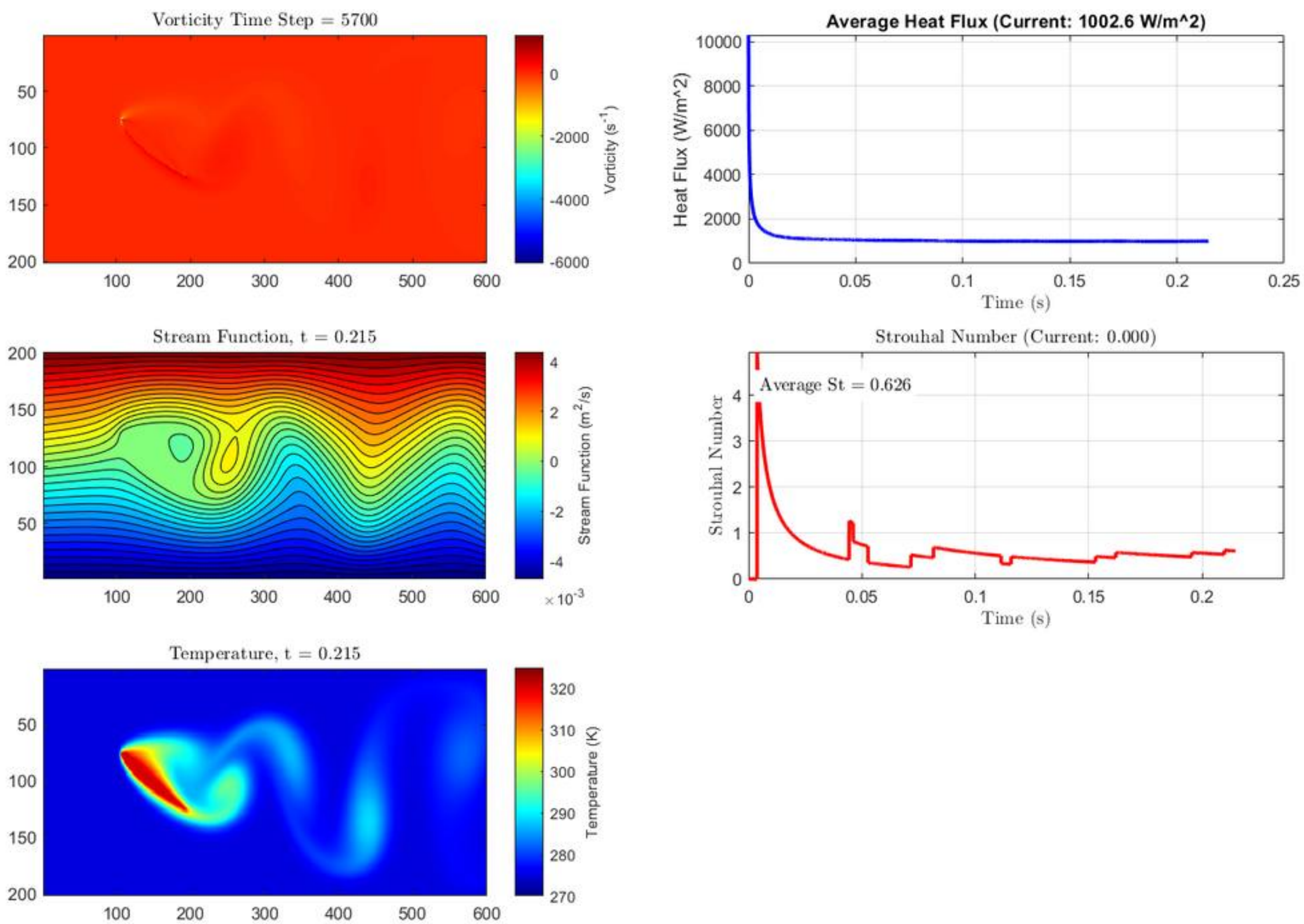


ATTACHED FLOW

Flow remains attached and the temperature distribution is smooth and symmetric about the chord line. The result was validated against SimFlow CFD — within 5% of the commercial result — confirming correct PDE discretization and boundary conditions.

Heat Transfer Simulation — NACA 2412 at 30° Angle of Attack

Apr – May 2025

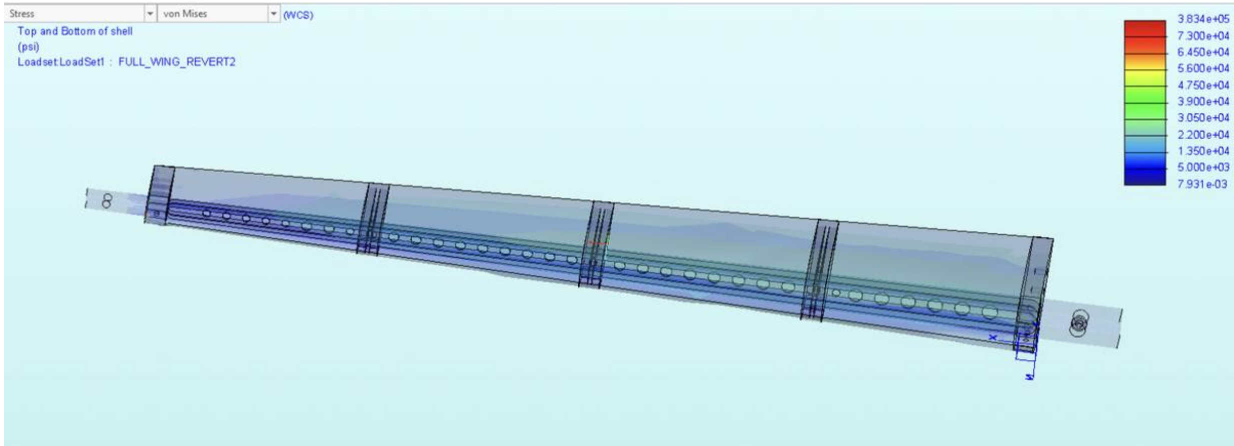


SEPARATED FLOW & VORTEX SHEDDING

Flow separates at the leading edge, producing large unsteady vortices shed periodically into the wake, with Strouhal number St = 0.62 — consistent with separated-flow theory. Higher stagnation temperatures and increased heat flux (13% above the 5° case) result from the thickened boundary layer.

Design and Manufacturing of a Load-Bearing Airfoil

Mar – May 2023



Full-wing FEA — von Mises stress at 97 lbf

Wing under 97 lbf test load

Completed wing assembly

OVERVIEW

The goal was to design a wing that could carry a specified load while minimizing deflection, cost, and weight. The 9-member team's 25.5-inch airfoil supported 97 lbf at 23.5 in. from the secured end with 3.75 in. tip deflection, weighed 18.3 oz, and failed at 120 lbs (beam web torsion) — above the 97 lbf design load.

DESIGN

- Aluminum 7075-T651 I-beam (yield stress 73,000 psi) with circular lightening holes — selected for best strength-to-weight ratio from four competing beam concepts.
- Nylon bulkheads with circular cutouts reduced weight.
- Balsa wood skin formed the NACA profile.

FEA & VALIDATION

- PTC Creo FEA validated deflection at 97 lbf.
- Failure mode — beam web torsion collapse at 120 lbs — was not captured in the planar FEA model, teaching the importance of 3D torsional load modeling.

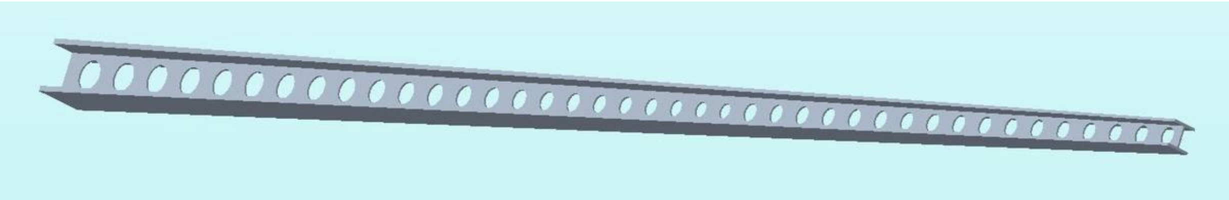
SKILLS USED

PTC Creo · FEA · CNC Milling (Haas VF-7/40) · Waterjet (ProtoMAX) · FDM & SLS 3D Printing · Structural Analysis · GD&T

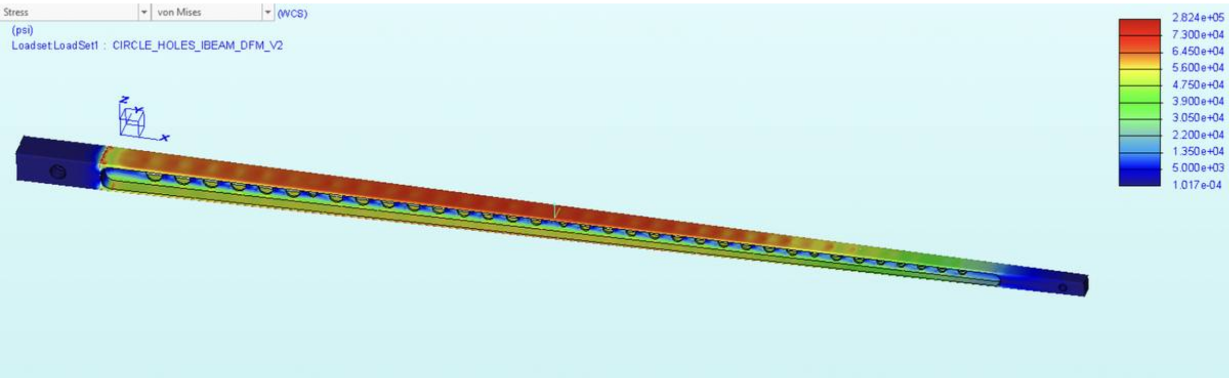
MECHANICAL ENGINEERING

Load-Bearing Airfoil — Manufacturing & Testing

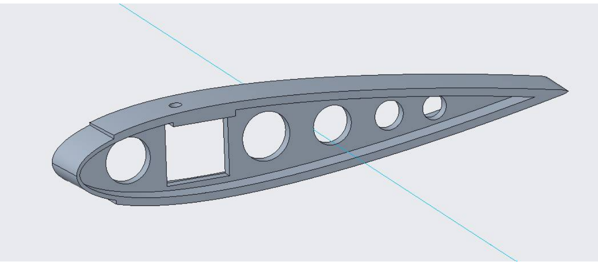
Mar – May 2023



Circular-Hole I-Beam



I-Beam FEA



Nylon Bulkhead Design

Waterjetted Bulkhead

MANUFACTURING

- I-beam CNC milled on Haas VF-7/40.
- Bulkheads waterjet-cut on ProtoMAX.
- Ribs printed on Ender 3 Pro (FDM) and structural parts on Formlabs SLS.
- Balsa skin cut on bandsaw and sanded to NACA profile.

MY CONTRIBUTION

Designed and performed FEA on the I-beam · Bulkhead material selection · Airfoil assembly

TESTING RESULTS

- Passed the 97 lbf design load at 3.75 in. deflection, then continued loading to failure at 120 lbs.
- All weight targets were met.

LESSONS LEARNED

- Torsional loads must be explicitly modeled in 3D FEA.
- Physical load paths in complex assemblies differ from simplified beam theory predictions, especially near joints and web openings.